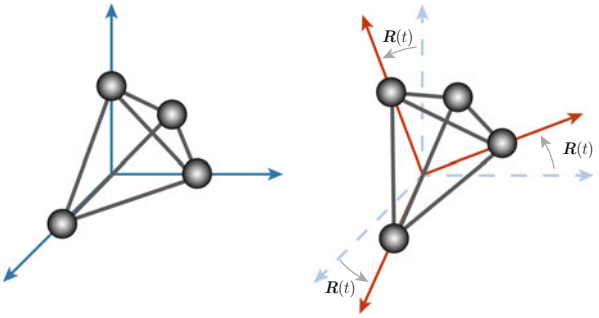


Fig. 4.7 A rigid rotation can be viewed as premultiplication by an orthogonal matrix $\boldsymbol{\Theta}$. In this way the orientation of the position vectors of points in the rigid body at any time t are described by a trajectory in the space of 3×3 orthogonal matrices, $\boldsymbol{\Theta}(t), t \in \mathbb{R}$

$\boldsymbol{\delta}_i(t) = \boldsymbol{\Theta}(t)\boldsymbol{\delta}_i(0)$, see Fig. 4.7, thus we have

$$K_{\text{rot}} = \frac{1}{2}\sum_{i=1}^{\text{N}} m_i \boldsymbol{\delta}_i(0)^T \dot{\boldsymbol{\Theta}}^T \dot{\boldsymbol{\Theta}} \boldsymbol{\delta}_i(0).$$

This expression may be rewritten as

$$K_{\text{rot}} = \frac{1}{2}\text{Tr}\left(\dot{\boldsymbol{\Theta}}\boldsymbol{R}\dot{\boldsymbol{\Theta}}^T\right),$$

where the fixed matrix $\boldsymbol{R}$ is given by

$$\boldsymbol{R} = \sum_{i=1}^{\text{N}} m_i \boldsymbol{\delta}_i(0)\boldsymbol{\delta}_i(0)^T.$$

By choosing an appropriate coordinate frame, it is possible to insure that the symmetric matrix $\boldsymbol{R}$ is a diagonal matrix.

An alternative form for the rotational kinetic energy may be obtained by writing

$$\begin{aligned}\dot{\boldsymbol{\delta}}_i &= \dot{\boldsymbol{\Theta}}\boldsymbol{\delta}_i(0) \\ &= \dot{\boldsymbol{\Theta}}\boldsymbol{\Theta}^T\boldsymbol{\Theta}\boldsymbol{\delta}_i(0) = \dot{\boldsymbol{\Theta}}\boldsymbol{\Theta}^T\dot{\boldsymbol{\delta}}_i,\end{aligned}$$

where we have used the fact that $\boldsymbol{\Theta}$ is orthogonal. Moreover, since $\boldsymbol{\Theta}\boldsymbol{\Theta}^T = \boldsymbol{I}$, we have

$$\dot{\boldsymbol{\Theta}}\boldsymbol{\Theta}^T + \boldsymbol{\Theta}\dot{\boldsymbol{\Theta}}^T = 0,$$

which indicates that $\dot{\boldsymbol{\Theta}}\boldsymbol{\Theta}^T$ is a skew symmetric matrix. Any 3×3 skew matrix may be written as

$$\dot{\boldsymbol{\Theta}}\boldsymbol{\Theta}^T = \text{skew}(\boldsymbol{\omega}) \equiv \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}. \tag{4.38}$$

$\boldsymbol{\omega}$ is referred to as the *spatial angular velocity* of the rigid body. It follows that

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\delta}_i = \text{skew}(\boldsymbol{\omega})\boldsymbol{\delta}_i = \boldsymbol{\omega} \times \boldsymbol{\delta}_i = -\text{skew}(\boldsymbol{\delta}_i)\boldsymbol{\omega}, \tag{4.39}$$

which is the usual definition.

We now insert the expression for $\dot{\boldsymbol{\delta}}_i$ into the rotational kinetic energy, to obtain

$$K_{\text{rot}} = \frac{1}{2}\sum_{i=1}^{\text{N}} m_i \|\dot{\boldsymbol{\delta}}_i\|^2 = -\frac{1}{2}\sum_{i=1}^{\text{N}} m_i \boldsymbol{\omega}^T \text{skew}(\boldsymbol{\delta}_i)^2 \boldsymbol{\omega}.$$

It is convenient to express this in terms of the so-called *body angular velocity* $\boldsymbol{\omega}_{\text{b}} = \boldsymbol{\Theta}^T\boldsymbol{\omega}$ in which case it becomes

$$K_{\text{rot}} = \frac{1}{2}\boldsymbol{\omega}_{\text{b}}^T \boldsymbol{T} \boldsymbol{\omega}_{\text{b}},$$

where the matrix $\boldsymbol{T} = -\sum_{i=1}^{\text{N}} m_i \text{skew}(\boldsymbol{\delta}_i(0))^2$ is referred to as the *moment of inertia tensor* or simply the *inertia tensor* and can be written as

$$\boldsymbol{T} = \sum_{i=1}^{\text{N}} m_i \left[\|\boldsymbol{\delta}_i(0)\|^2 \boldsymbol{I} - \boldsymbol{\delta}_i(0)\boldsymbol{\delta}_i(0)^T \right].$$

This matrix is closely related to the matrix $\boldsymbol{R}$:

$$\boldsymbol{R} = \sum_{i=1}^{\text{N}} m_i \boldsymbol{\delta}_i \boldsymbol{\delta}_i^T = \sum_{i=1}^{\text{N}} m_i \|\delta_i\|^2 \boldsymbol{I} - \boldsymbol{T} = (T_{11} + T_{22} + T_{33})\boldsymbol{I} - \boldsymbol{T}. \tag{4.40}$$

Example 4.1 (Dipolar Molecular Liquid) The Stockmayer model for a molecular liquid with permanent dipoles [388] treats this as a system of rigid bodies (spheres) with dipole-dipole interaction potential; we already encountered this in the exercises of Chap. 1. From a computational and modelling perspective, there is no real reason to use spheres (and perhaps good reasons not to); for example they might be taken to be ellipsoidal. Let us assume that they are identical ellipsoids. The Hamiltonian for a more general dipolar system can be written

$$\begin{aligned} H = {} & K_{\text{rot}}(\boldsymbol{\pi}_1, \boldsymbol{\pi}_2, \ldots, \boldsymbol{\pi}_{\text{N}}) + K_{\text{trans}}(\boldsymbol{p}_1, \boldsymbol{p}_2, \ldots, \boldsymbol{p}_{\text{N}}) \\ & + U(\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_{\text{N}}, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \ldots, \boldsymbol{\mu}_{\text{N}}), \end{aligned}$$